BT3041 Analysis and Interpretation of Biological Data · Term project

What does tumour gene expression reveal about the clinical receptor groups of breast cancer?

TCGA breast cancer RNA-seq (1,088 tumours), checked against protein (RPPA) in the same tumours and an independent cohort (METABRIC, 1,980 tumours)

HR+/HER2−
n = 591

HER2+
n = 160

TNBC
n = 157

Groups from clinical IHC/FISH tests (ER, PR, HER2), not from expression

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IIT Madras · November 2026 · Code: github.com/Vijayavallabh/BT3041_Project

Receptor tests guide treatment. What does expression add?

Background

• Pathologists test ER and PR (IHC) and HER2 (IHC/FISH); the result picks endocrine therapy, anti-HER2 drugs or chemotherapy.

• TNBC (all three negative) has no targeted therapy and relapses early.

• PAM50 subtypes are computed from expression itself, so "does expression separate PAM50?" is circular. We use the clinical labels and keep PAM50 only as a comparator.

Research questions

Q1 Does label-free structure (PCA, UMAP, clustering) recover the receptor groups? slides 4–5

Q2 Which genes and pathways differ between TNBC and HR+/HER2− (DE + FDR, GSEA)? slides 6–7

Q3 Can expression predict clinical ER and HER2 status better than the receptor gene alone? slide 8

Validation design (slide 9): every result is checked outside the data it came from

Labels

Clinical IHC/FISH for ER, PR, HER2. PAM50 only as a comparator; PAM50 genes left out of clustering and ablations.

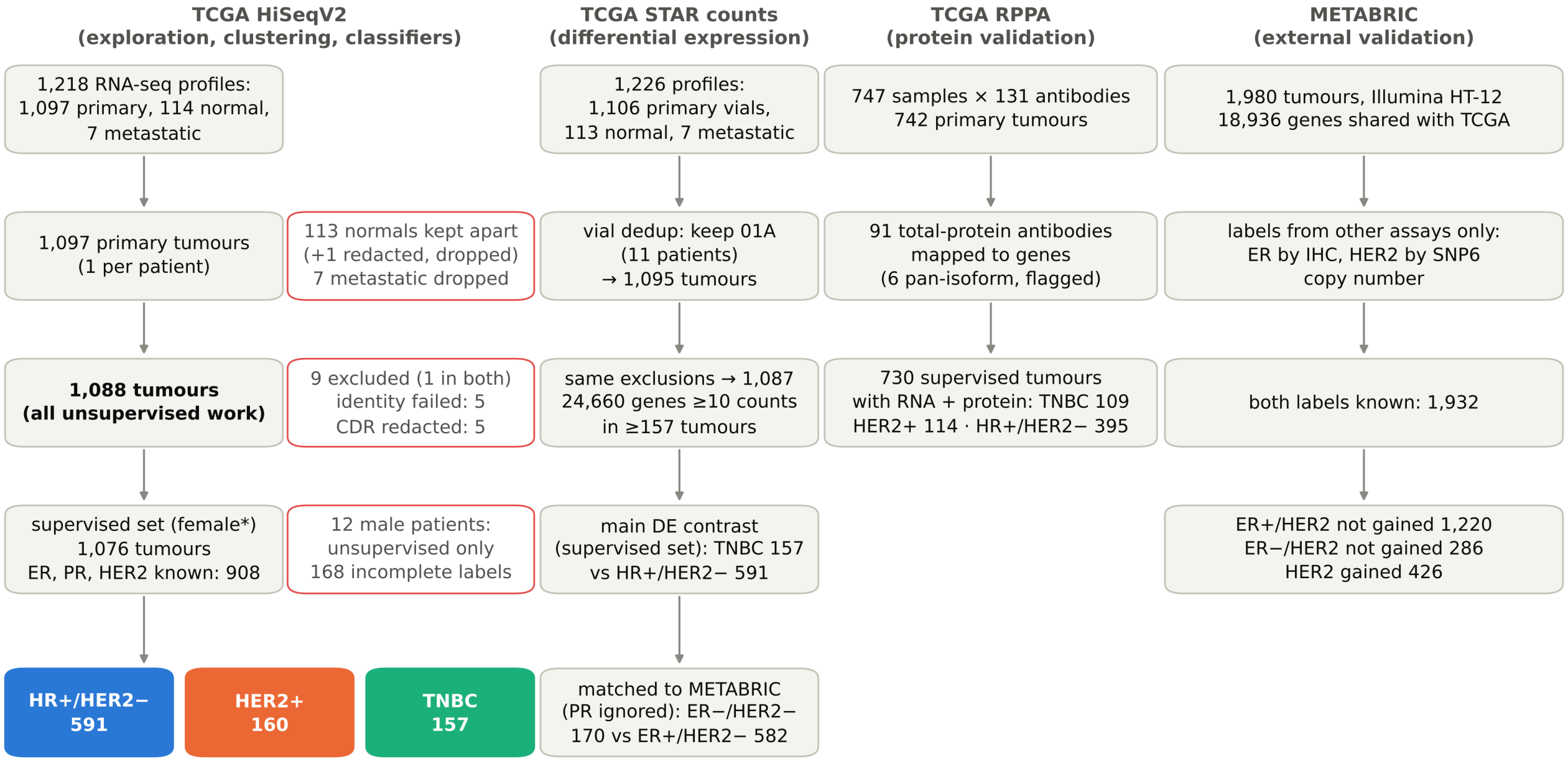
Protein

TCGA RPPA on the same tumours (730 with RNA and protein): do mRNA changes show up in protein?

Independent cohort

METABRIC microarrays: ER by IHC, HER2 by DNA copy number. Models frozen on TCGA before any METABRIC label was read.

Sample flow: from downloaded files to analysis groups

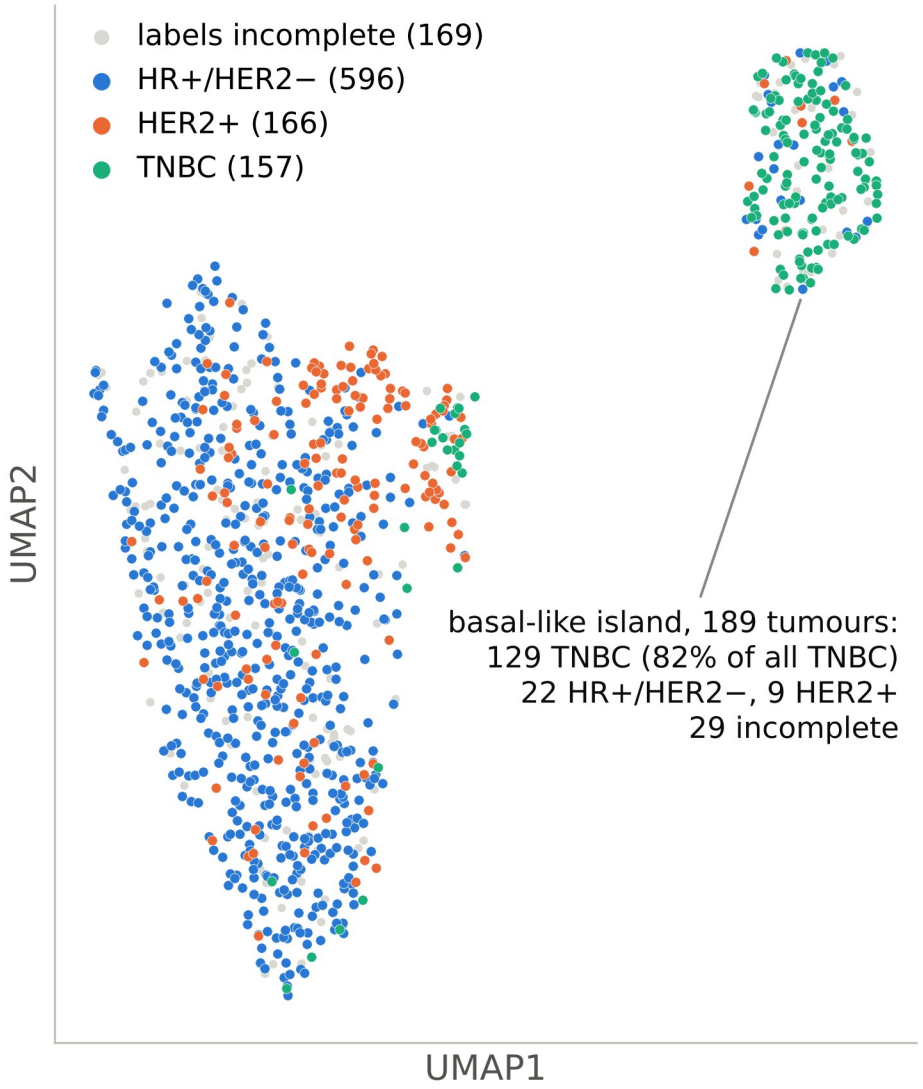


Expression sets TNBC apart: a basal-like island holds 82% of TNBC.

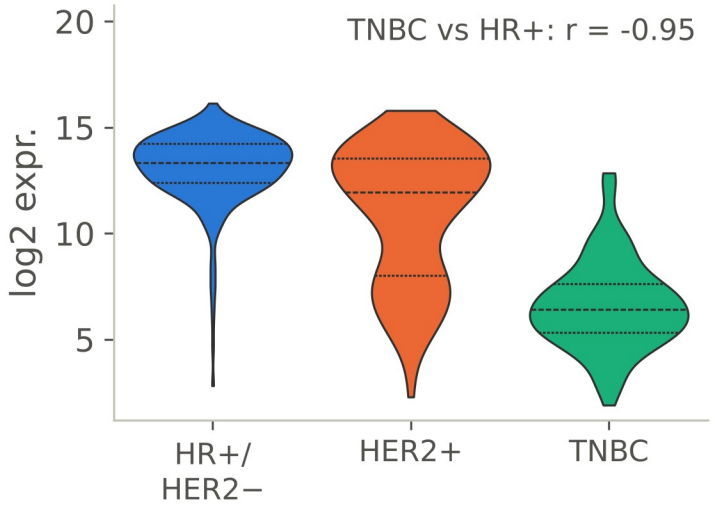
HER2+ tumours mostly overlap HR+/HER2− tumours; their clearest difference is ERBB2.

Violins: female supervised set (HR+/HER2− 591, HER2+ 160, TNBC 157); r = rank-biserial effect size; all four q < 1e-10 (Mann-Whitney, BH over 16 tests)

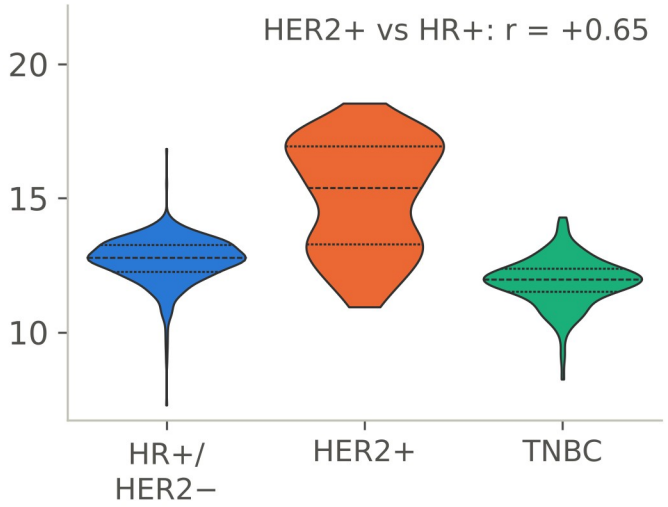
UMAP of 1,088 tumours (top 2,000 genes)



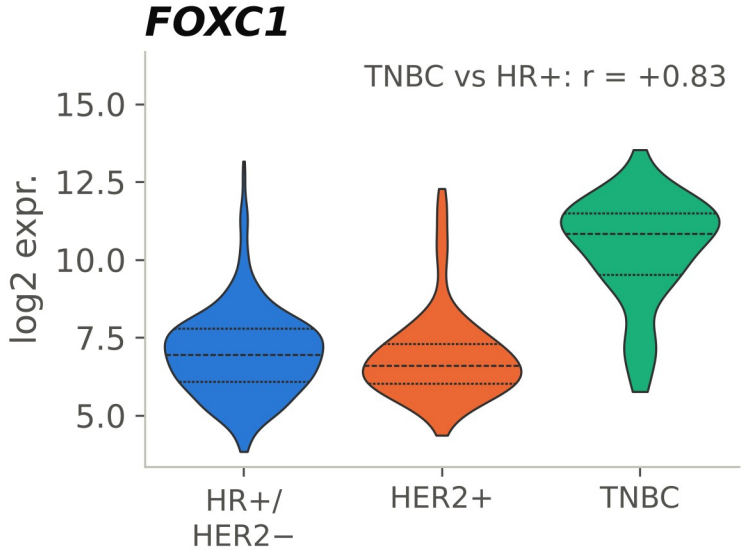
ESR1



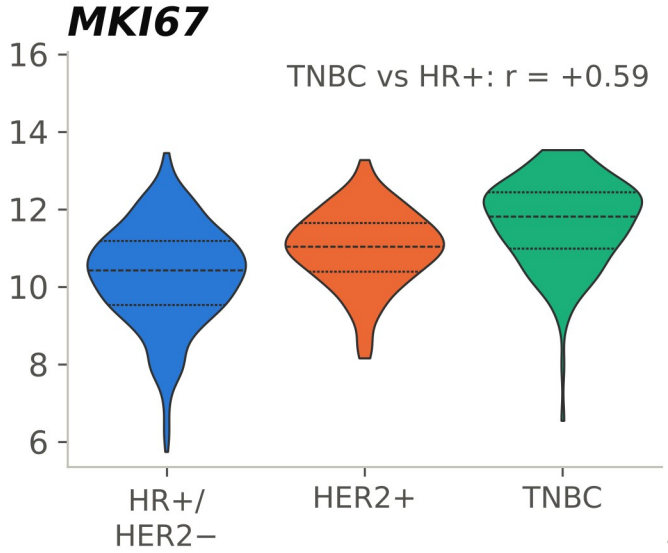
ERBB2



FOXC1



MKI67

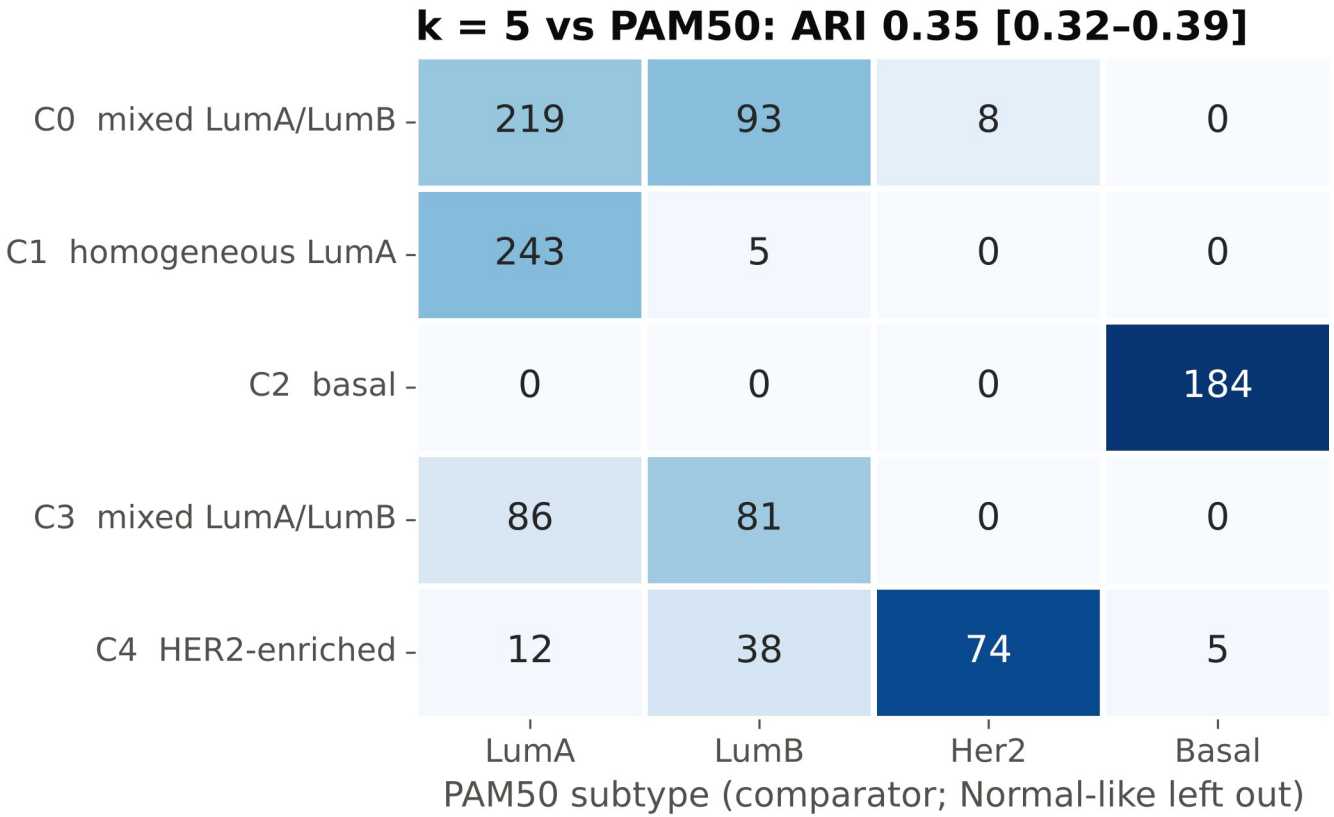


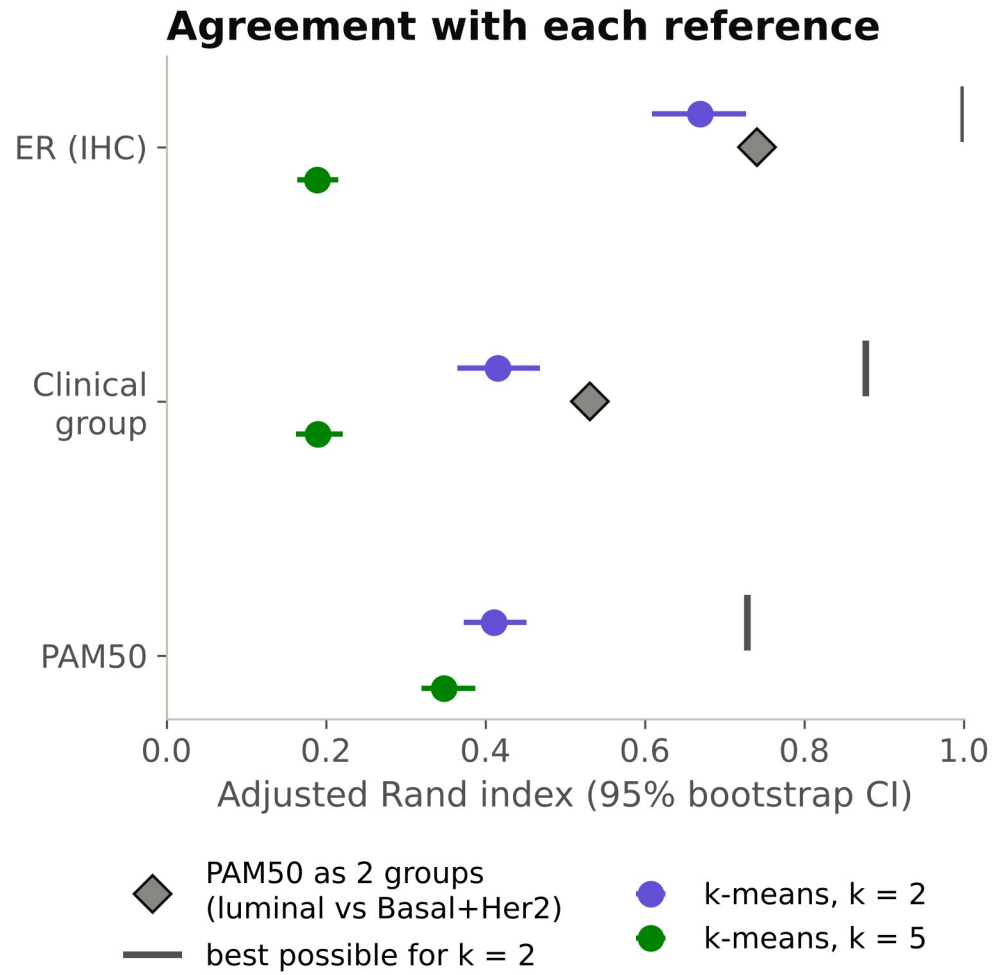
Label-free clustering is most stable at k = 2: a basal-like group of 217 tumours splits off

It holds 188/189 PAM50 Basal, 138/157 TNBC and 179/235 ER-negative tumours

Finer clusters (k = 5) also recover a HER2-enriched group but split luminal tumours differently from PAM50

1,088 tumours, 2,000 most variable genes, PAM50 genes left out. k = 5 is less stable than k = 2 (PAC 0.14 vs 0.02; ARI 0.67 against a 5,000-gene rerun).





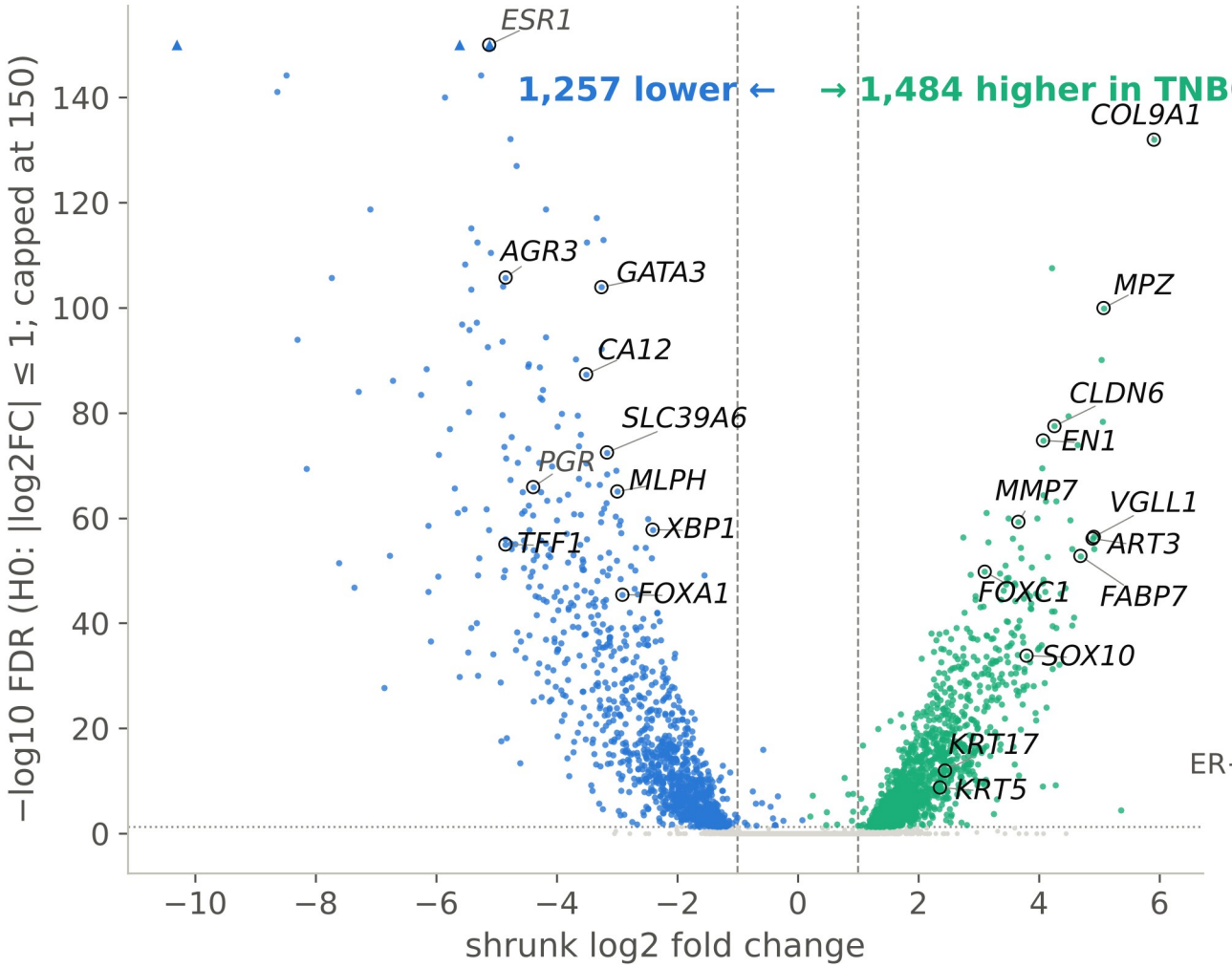
Heatmap: cell = count, shade = share of each PAM50 column. Row names: closest Netanely 2016 K = 5 cluster by majority PAM50 class (a heuristic, not their assignment; they also clustered adjacent normals).

2,741 genes differ more than 2-fold between TNBC and HR+/HER2− tumours (FDR 5%).

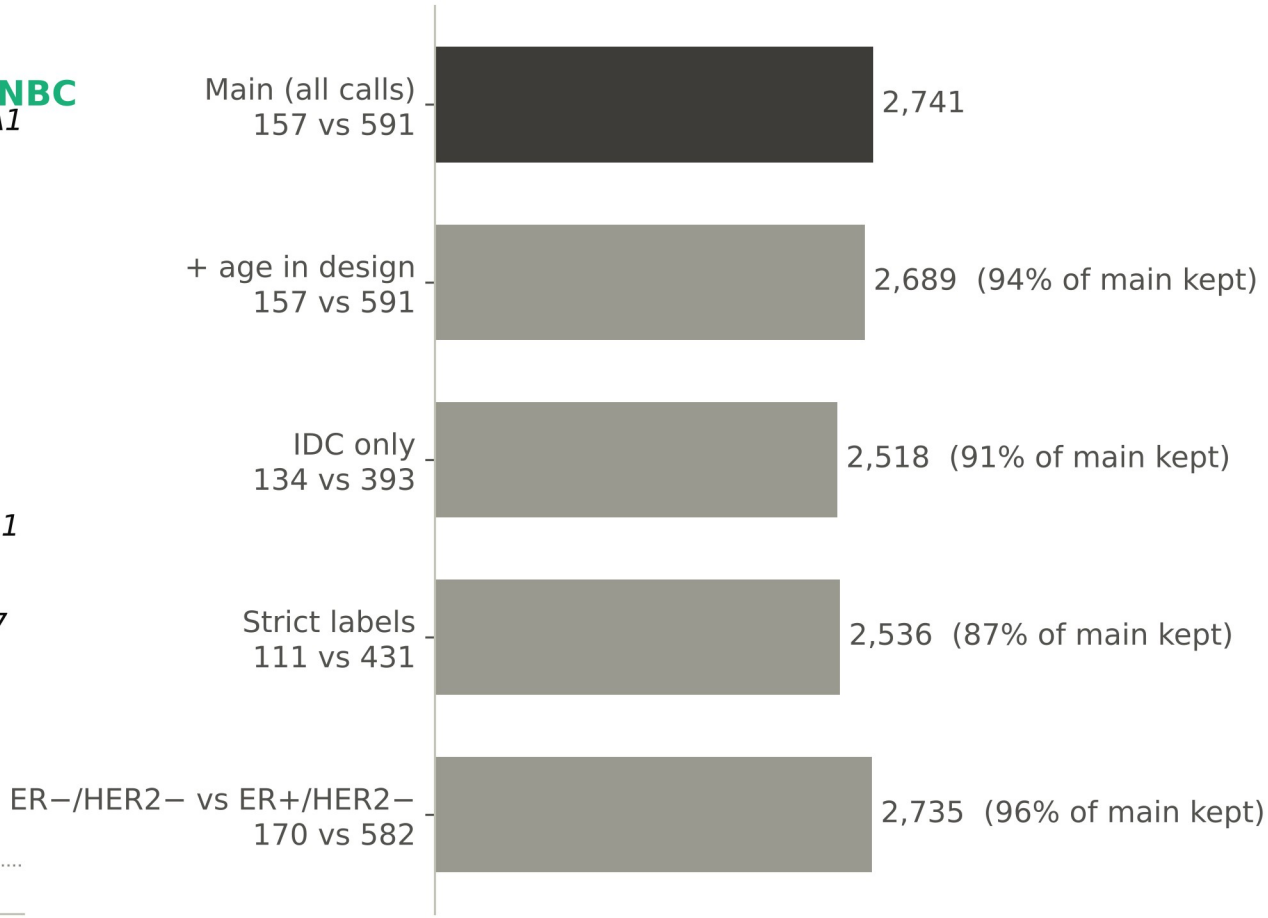
Robust to dropping lobular tumours (91% kept), adding age (94%) and strict labels (87%).

The 2-fold threshold is tested (lfc_null = 1), not applied afterwards: a plain Wald test calls 18,604 of 23,439 genes (79%).
ESR1 and PGR are positive controls; beyond them: basal transcription factors (FOXC1, EN1, SOX10) up, the ER regulon (GATA3, FOXA1, XBP1, AGR3) down.

TNBC vs HR+/HER2− (PyDESeq2, ~ histology + group)



DEGs in each sensitivity run



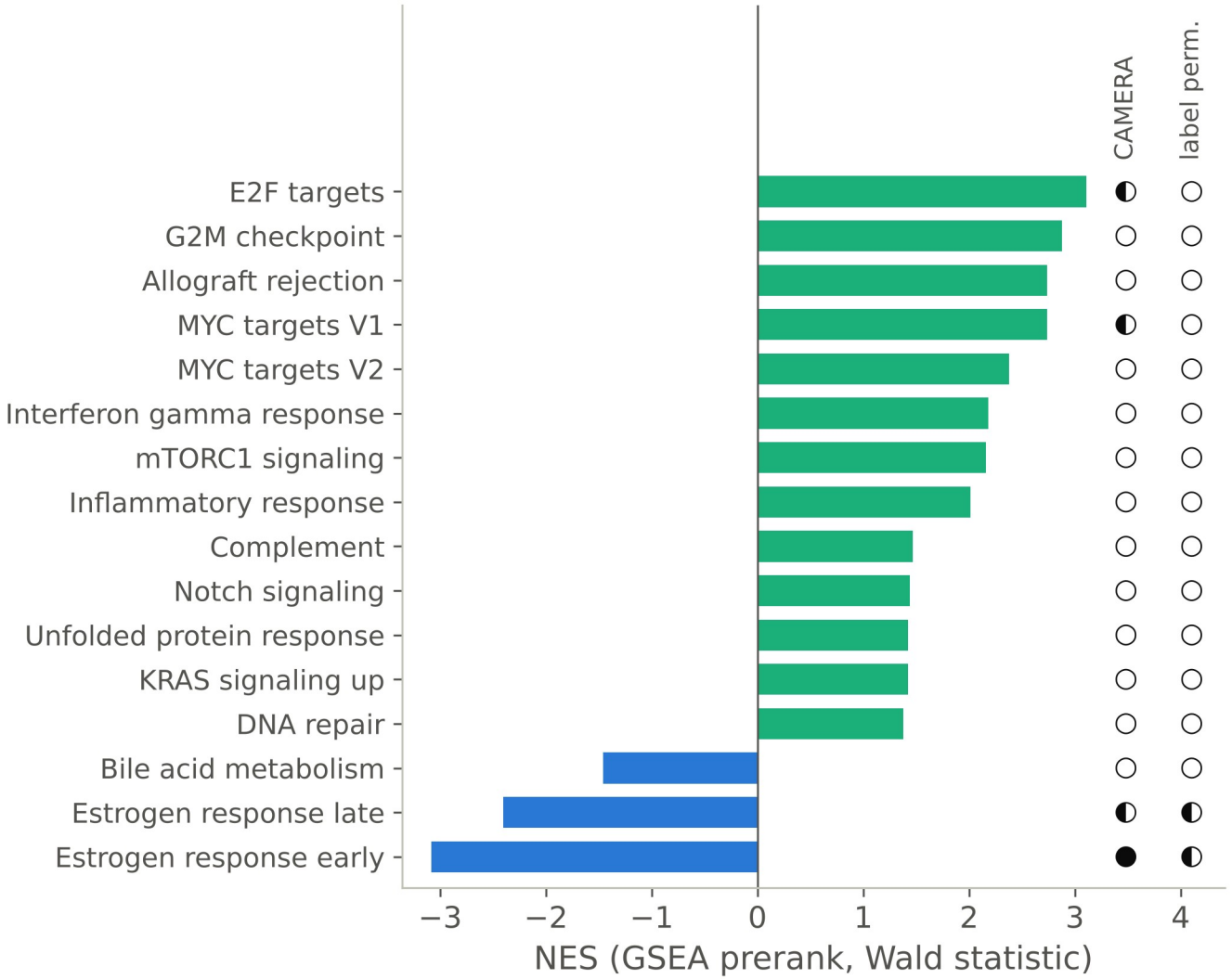
STAR counts, female supervised set; shrunk LFC (apeGLM-type prior); HER2+ vs HR+/HER2−: 148 DEGs, 33 of them within 2 Mb of ERBB2 (17q12 amplicon).

TNBC ranks higher on proliferation, MYC and immune programmes, lower on estrogen response.

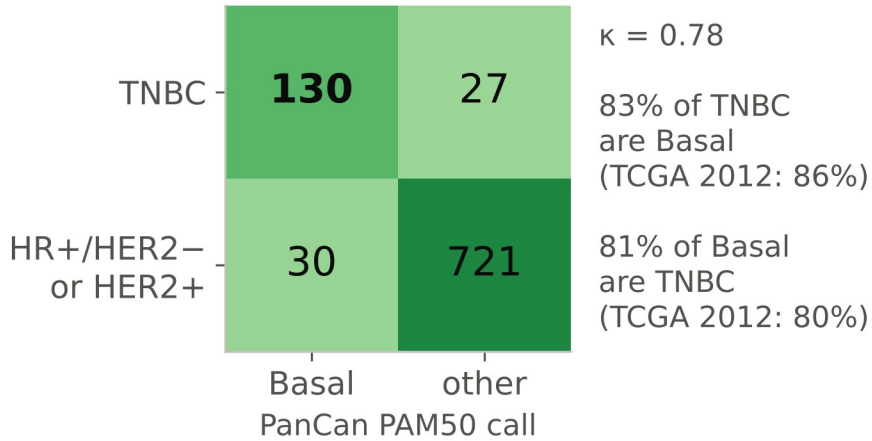
Correlation-aware checks agree on direction; only early estrogen response stays significant.

NES: GSEA prerank on the DESeq2 Wald statistic (gene-permutation q-values are optimistic when genes are co-expressed).
Marks: ● q < 0.05, ◐ nominal p < 0.05, ○ neither, for CAMERA (competitive test with inter-gene correlation) and label-permutation GSEA.

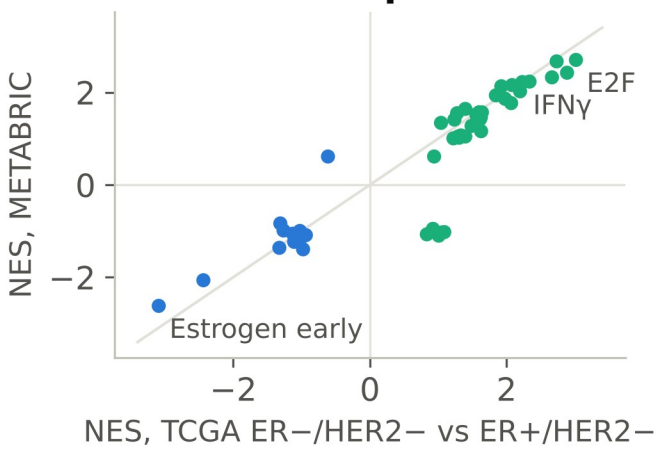
Hallmarks at prerank FDR < 0.05 (16 of 28 shown)



Clinical TNBC vs PAM50 Basal



METABRIC replicates: NES ρ = 0.94

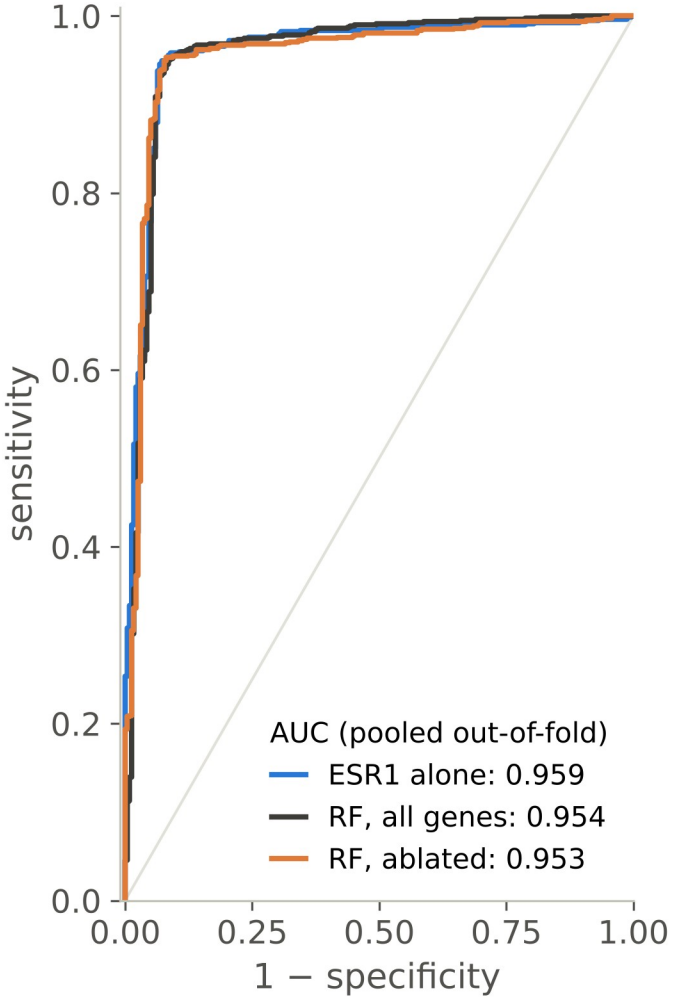


Models match but do not beat the receptor gene alone (ESR1 0.96, ERBB2 0.85).

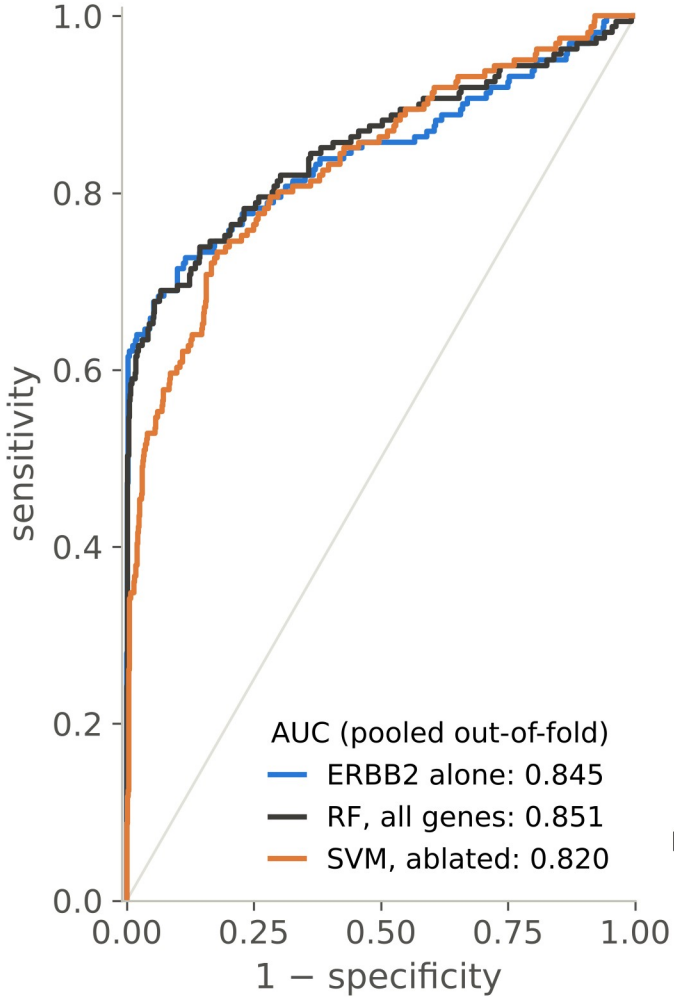
Without receptor genes, 17q12 amplicon and PAM50: ER unchanged (0.95), HER2 down to 0.78–0.82.

Best model vs gene alone, Nadeau-Bengio corrected t on 15 paired folds: ER RF ΔAUC -0.004 (p 0.69); HER2 RF ΔAUC +0.008 (p 0.66).
Shuffled labels: MCC -0.01 (ER), +0.04 (HER2). Filtering, scaling and feature selection refitted inside every training fold.

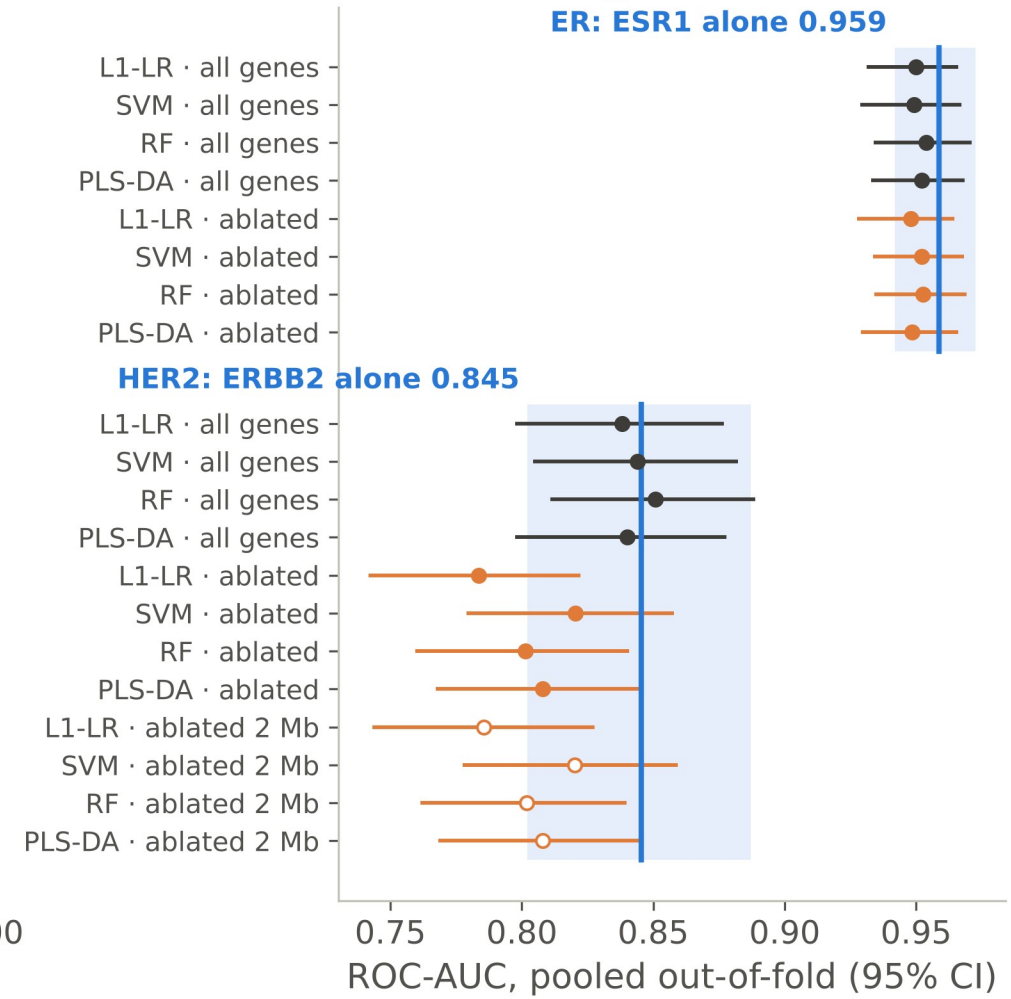
Clinical ER (IHC)



Clinical HER2 (IHC/FISH)



4 models × gene sets



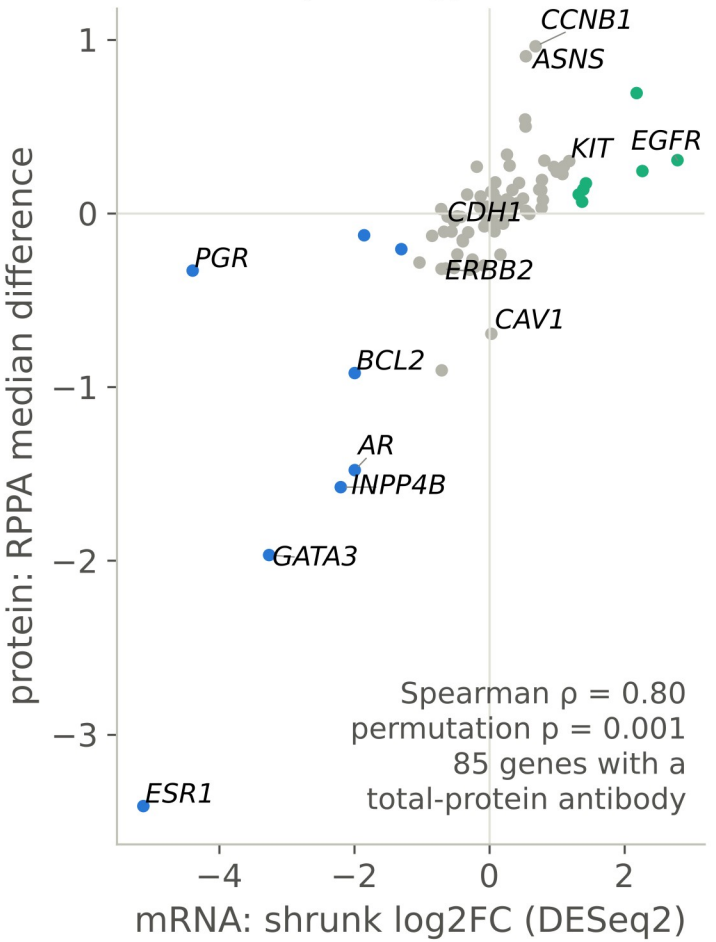
ER 789+/235−, HER2 161+/752−. Nested CV: outer 5-fold × 3, inner 5-fold. Ablated: minus ESR1/PGR/ERBB2, genes ±1.5 Mb (or 2 Mb) of ERBB2, and PAM50.

Holds in protein and in METABRIC: ER AUC 0.96, HER2 0.91, ER−/HER2− 0.93.

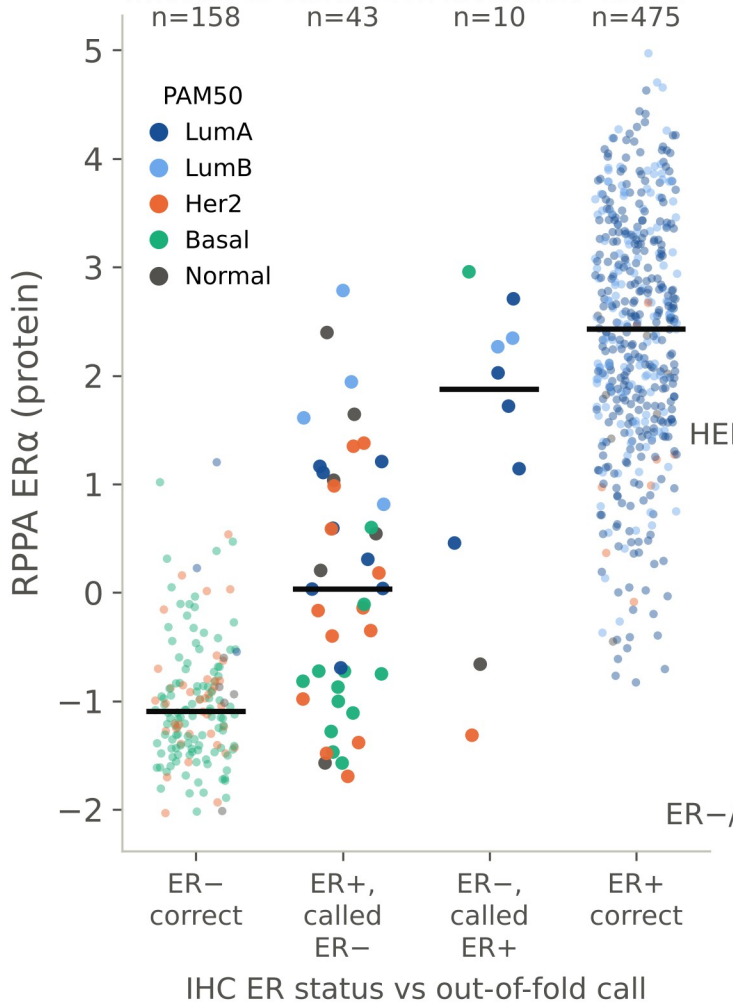
mRNA effects match protein (ρ = 0.80); 86% of TCGA DEGs replicate in METABRIC.

IHC ER− tumours called ER+ have ER+-level ERα protein (8/10 above the ER− 95th percentile): likely IHC false negatives. DEGs: same sign, q < 0.05.
METABRIC labels: ER by IHC, HER2 by SNP6 copy number; models frozen on TCGA before any METABRIC label was read.
HER2 transfers better than it cross-validates: copy-number labels track mRNA more closely than IHC/FISH calls.

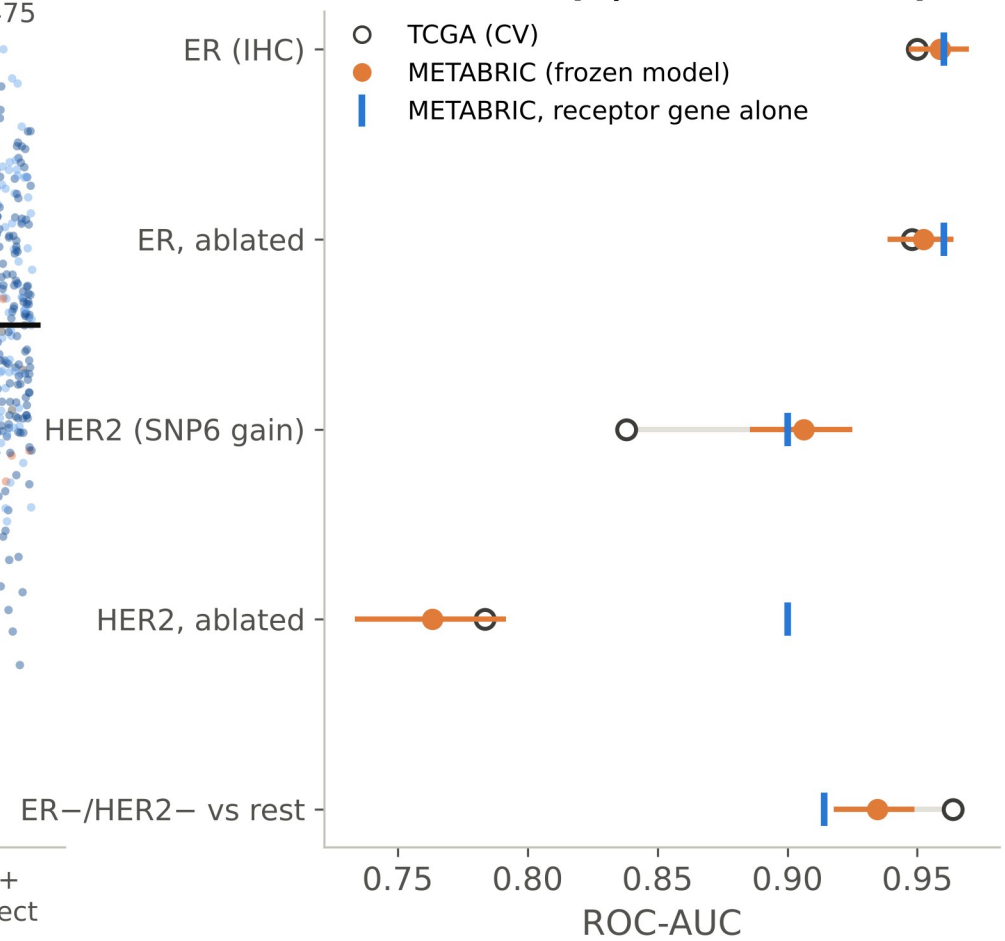
Protein (RPPA), same tumours



Where the ER model errs



METABRIC (1,980 tumours)



Conclusions

1. TNBC is a distinct transcriptional state

A basal-like island holds 82% of TNBC; 2,741 genes differ > 2-fold: ER regulon down, basal factors, proliferation and MYC up. HER2+ differs far less (148 genes, led by the 17q12 amplicon).

2. The receptor gene already says almost everything

Models match but do not beat ESR1 (0.96) and ERBB2 (0.85). ER survives removing ESR1 (0.95): a regulon. HER2 falls to 0.78–0.82 without the amplicon.

3. Findings hold beyond the training data

mRNA matches protein (ρ = 0.80); METABRIC transfer AUC 0.96 (ER), 0.91 (HER2); 86% of DEGs replicate. Several model 'errors' look like IHC false negatives.

Limitations

- Pathway q-values: only early estrogen response survives correlation-aware tests.
- DE not adjusted for tumour purity; proliferation and immune shifts are partly composition.
- HER2 labels from status text alone are noisy (they cost about 0.05 AUC).
- TCGA follow-up is short: survival is exploratory (100 progression events).

Done now / for the final report

Done: data preparation, exploration, clustering, DE, pathways, classification, RPPA and METABRIC validation.

Report: survival in TCGA and METABRIC, label-rule sensitivity, comparison with 21 published numbers.

Future: purity-adjusted DE; proteomic (CPTAC) labels.

Code: github.com/Vijayavallabh/BT3041_Project